1.Write a C program to find the maximum and minimum elements in an array

```c
#include <stdio.h>

int main() {
   int arr[] = {12, 45, 2, 67, 34, 89, 5, 23};
   int n = sizeof(arr) / sizeof(arr[0]);

   int max = arr[0];
   int min = arr[0];

   for (int i = 1; i < n; i++) {
       if (arr[i] > max) {
           max = arr[i];
       } else if (arr[i] < min) {
           min = arr[i];
       }
   }

   printf("Array elements: ");
   for(int i = 0; i < n; i++) printf("%d ", arr[i]);

   printf("\n\nMaximum element: %d", max);
   printf("\nMinimum element: %d\n", min);

   return 0;
}
```

Output

```
/Users/Dolkar/.zprofile:10: no such file or directory: /opt/homebrew/bin/brew
tenzinlhamo@TENTENs-MacBook-Pro CProgramming % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q1"
Array elements: 12 45 2 67 34 89 5 23

Maximum element: 89
Minimum element: 2
```

2.Implement a C program to reverse the elements of an array.

```c
#include <stdio.h>
int main() {
   int arr[] = {10, 20, 30, 40, 50};
   int n = sizeof(arr) / sizeof(arr[0]);
   int start = 0;
   int end = n - 1;
   int temp;
```

```
    while (start < end) {
        temp = arr[start];
        arr[start] = arr[end];
        arr[end] = temp;

        start++;
        end--;
    }

    printf("Reversed array: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

Output

```
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q2"
Reversed array: 50 40 30 20 10
tenzinlhamo@TENTENs-MacBook-Pro output %
```

3. Write a C program to sort an array of integers in ascending order using a sorting algorithm of your choice (e.g., bubble sort, selection sort, insertion sort).

Code

```c
#include <stdio.h>
int main() {
    int arr[] = {64, 34, 25, 12, 22, 11, 90};
    int n = sizeof(arr) / sizeof(arr[0]);
    int temp;

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    printf("Sorted array: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolk
./"q3"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q3"
Sorted array: 11 12 22 25 34 64 90
tenzinlhamo@TENTENs-MacBook-Pro output %
```

4.Implement a C program to find the second largest element in an array.

Code

```c
#include <stdio.h>
#include <limits.h>

int main() {
   int arr[] = {12, 35, 1, 10, 34, 1};
   int n = sizeof(arr) / sizeof(arr[0]);

   if (n < 2) {
       printf("Array size must be at least 2.\n");
       return 1;
   }

   int first = INT_MIN;
   int second = INT_MIN;

   for (int i = 0; i < n; i++) {
       if (arr[i] > first) {
           second = first;
           first = arr[i];
       } else if (arr[i] > second && arr[i] != first) {
           second = arr[i];
       }
   }

   if (second == INT_MIN) {
       printf("No second largest element found.\n");
   } else {
       printf("The second largest element is %d\n", second);
   }

   return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q4"
The second largest element is 34
tenzinlhamo@TENTENs-MacBook-Pro output %
```

5.Write a C program to merge two sorted arrays into a single sorted array.

Code

```
include <stdio.h>
void mergeArrays(int arr1[], int n1, int arr2[], int n2, int merged[]) {
   int i = 0, j = 0, k = 0;

   while (i < n1 && j < n2) {
       if (arr1[i] < arr2[j]) {
           merged[k++] = arr1[i++];
       } else {
           merged[k++] = arr2[j++];
       }
   }

   while (i < n1) {
       merged[k++] = arr1[i++];
   }

   while (j < n2) {
       merged[k++] = arr2[j++];
   }
}

int main() {
   int arr1[] = {1, 3, 5, 7};
   int n1 = sizeof(arr1) / sizeof(arr1[0]);
   int arr2[] = {2, 4, 6, 8};
   int n2 = sizeof(arr2) / sizeof(arr2[0]);
   int merged[n1 + n2];
   mergeArrays(arr1, n1, arr2, n2, merged);
   printf("Merged array: ");
   for (int i = 0; i < n1 + n2; i++) {
       printf("%d ", merged[i]);
   }
   printf("\n");
   return 0;
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q5"
Merged array: 1 2 3 4 5 6 7 8
tenzinlhamo@TENTENs-MacBook-Pro output %
```

6. Write a C program to find the length of a string without using the built-in string functions.

Code

```c
#include <stdio.h>
int main() {
   char str[100];
   int length = 0;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);

   while (str[length] != '\0') {
       length++;
   }

   if (length > 0 && str[length - 1] == '\n') {
       length--;
   }

   printf("The length of the string is: %d\n", length);

   return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q6"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q6"
Enter a string: 5
The length of the string is: 1
tenzinlhamo@TENTENs-MacBook-Pro output %
```

7.Implement a C program to reverse a string.

Code

```c
#include <stdio.h>
int main() {
   char str[100];
   int start = 0, end = 0;
   char temp;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);
   while (str[end] != '\0' && str[end] != '\n') {
       end++;
   }
   end--;
   while (start < end) {
       temp = str[start];
       str[start] = str[end];
       str[end] = temp;
       start++;
       end--;
   }

   printf("Reversed string: %s\n", str);

   return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q7"
Enter a string: 2 3 4 5 6
Reversed string: 6 5 4 3 2
```

8.Write a C program to check if a given string is a palindrome

Code

```c
#include <stdio.h>
#include <stdbool.h>

int main() {
   char str[100];
   int length = 0;
```

```c
    bool isPalindrome = true;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    while (str[length] != '\0' && str[length] != '\n') {
        length++;
    }
    for (int i = 0; i < length / 2; i++) {
        if (str[i] != str[length - 1 - i]) {
            isPalindrome = false;
            break;
        }
    }

    if (isPalindrome) {
        printf("The string is a palindrome.\n");
    } else {
        printf("The string is not a palindrome.\n");
    }

    return 0;
```

Output

```
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q8"
Enter a string: 9 8 3 2 3 4 2
The string is not a palindrome.
tenzinlhamo@TENTENs-MacBook-Pro output %
```

9.Implement a C program to count the occurrence of a specific character in a string.

Code

```c
#include <stdio.h>
int main() {
    char str[100];
    char target;
    int count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
```

```c
    printf("Enter a character to search for: ");
    scanf("%c", &target);

    for (int i = 0; str[i] != '\0' && str[i] != '\n'; i++) {
        if (str[i] == target) {
            count++;
        }
    }

    printf("The character '%c' appears %d times.\n", target, count);

    return 0;
}
```

Output

```
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q8"
Enter a string: 9 8 3 2 3 4 2
The string is not a palindrome.
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q9"
Enter a string: 3 4 2 3 4
Enter a character to search for: 2
The character '2' appears 1 times.
tenzinlhamo@TENTENs-MacBook-Pro output %
```

10.Write a C program to concatenate two strings without using the built-in string functions.

Code

```c
#include <stdio.h>
int main() {
    char str1[100] = "Hello ";
    char str2[] = "World!";
    int i = 0, j = 0;

    while (str1[i] != '\0') {
        i++;
    }

    while (str2[j] != '\0') {
        str1[i] = str2[j];
        i++;
        j++;
```

```
    }

    str1[i] = '\0';
    printf("Concatenated string: %s\n", str1);

    return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q10"
Concatenated string: Hello World!
tenzinlhamo@TENTENs-MacBook-Pro output %
```

11.Write a function named calculateAverage that takes an array of integers as input and returns the average of the numbers.

Code

```
#include <stdio.h>
double calculateAverage(int arr[], int size) {
    if (size <= 0) {
        return 0.0;
    }

    int sum = 0;
    for (int i = 0; i < size; i++) {
        sum += arr[i];
    }

    return (double)sum / size;
}

int main() {
    int numbers[] = {10, 20, 30, 40, 50};
    int n = sizeof(numbers) / sizeof(numbers[0]);

    double avg = calculateAverage(numbers, n);

    printf("The average is: %.2f\n", avg);

    return 0;
```

```
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q11"
The average is: 30.00
tenzinlhamo@TENTENs-MacBook-Pro output %
```

12. Write a function named isPalindrome that takes a string as input and returns 1 if it is a palindrome (reads the same forwards and backwards), and 0 otherwise.

Code

```c
#include <stdio.h>
int isPalindrome(char str[]) {
   int length = 0;
   while (str[length] != '\0' && str[length] != '\n') {
       length++;
   }
   for (int i = 0; i < length / 2; i++) {
       if (str[i] != str[length - 1 - i]) {
           return 0;
       }
   }
   return 1;
}
int main() {
   char input[100];
   printf("Enter a string: ");
   fgets(input, sizeof(input), stdin);
   if (isPalindrome(input)) {
       printf("1 (It is a palindrome)\n");
   } else {
       printf("0 (It is not a palindrome)\n");
   }
   return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q12"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q12"
Enter a string: 8 3 2 4 6 7
0 (It is not a palindrome)
tenzinlhamo@TENTENs-MacBook-Pro output %
```

13.Write a function named findFactorial that takes an integer as input and returns its factorial.

Code

```c
#include <stdio.h>
long long findFactorial(int n) {
   if (n < 0) {
       return -1;
   }

   long long factorial = 1;
   for (int i = 1; i <= n; i++) {
       factorial *= i;
   }

   return factorial;
}

int main() {
   int num;
   printf("Enter a positive integer: ");
   scanf("%d", &num);

   long long result = findFactorial(num);

   if (result == -1) {
       printf("Factorial of negative numbers is undefined.\n");
   } else {
       printf("Factorial of %d is %lld\n", num, result);
   }

   return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q13"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q13"
Enter a positive integer: 8
Factorial of 8 is 40320
tenzinlhamo@TENTENs-MacBook-Pro output %
```

14.Write a function named convertTemperature that takes a temperature value in Celsius and converts it to Fahrenheit. The function should return the converted temperature.

Code

```c
#include <stdio.h>

float convertTemperature(float celsius) {
   return (celsius * 9 / 5) + 32;
}

int main() {
   float celsius;

   printf("Enter temperature in Celsius: ");
   scanf("%f", &celsius);

   float fahrenheit = convertTemperature(celsius);

   printf("%.2f Celsius is equal to %.2f Fahrenheit\n", celsius, fahrenheit);

   return 0;
}
```

Output

```
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q14"
Enter temperature in Celsius: 38
38.00 Celsius is equal to 100.40 Fahrenheit
tenzinlhamo@TENTENs-MacBook-Pro output %
```

15.Write a function named countOccurrences that takes a string and a character as input and returns the number of times the character appears in the string.

Code

```c
#include <stdio.h>
int countOccurrences(char str[], char target) {
   int count = 0;

   for (int i = 0; str[i] != '\0'; i++) {
       if (str[i] == target) {
           count++;
       }
   }
```

```
    return count;
}

int main() {
    char text[] = "programming in c is fun!";
    char search = 'n';

    int result = countOccurrences(text, search);

    printf("The character '%c' appears %d times in the string.\n", search, result);

    return 0;
}
```

Output

```
● tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
  ./"q15"
● tenzinlhamo@TENTENs-MacBook-Pro output % ./"q15"
  The character 'n' appears 3 times in the string.
```

16.Write a function named reverseArray that takes an array of integers as input and reverses the order of the elements in the array.

Code

```
#include <stdio.h>
void reverseArray(int arr[], int size) {
    int start = 0;
    int end = size - 1;
    int temp;

    while (start < end) {
        temp = arr[start];
        arr[start] = arr[end];
        arr[end] = temp;

        start++;
        end--;
    }
}
int main() {
    int numbers[] = {1, 2, 3, 4, 5};
```

```
    int n = sizeof(numbers) / sizeof(numbers[0]);

    reverseArray(numbers, n);

    printf("Reversed array: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", numbers[i]);
    }
    printf("\n");
    return 0;
}
```

Output

```
TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q16"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q16"
Reversed array: 5 4 3 2 1
tenzinlhamo@TENTENs-MacBook-Pro output %
```

17. Write a function named calculatePower that takes two integers, base and exponent, as input and returns the result of raising the base to the exponent.

Code

```
#include <stdio.h>
long long calculatePower(int base, int exponent) {
    long long result = 1;
    if (exponent == 0) {
        return 1;
    }
    for (int i = 0; i < exponent; i++) {
        result *= base;
    }

    return result;
}

int main() {
    int b, e;

    printf("Enter base: ");
    scanf("%d", &b);
    printf("Enter exponent: ");
```

```
    scanf("%d", &e);

    long long power = calculatePower(b, e);

    printf("%d raised to the power of %d is %lld\n", b, e, power);

    return 0;
}
```

Output

```
TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q17"
Enter base: 4
Enter exponent: 6
4 raised to the power of 6 is 4096
tenzinlhamo@TENTENs-MacBook-Pro output %
```

18.Write a function named findPrimeNumbers that takes an integer n as input and prints all prime numbers from 1 to n.

Code

```
#include <stdio.h>
void findPrimeNumbers(int n) {
    for (int num = 2; num <= n; num++) {
        int isPrime = 1;

        for (int i = 2; i * i <= num; i++) {
            if (num % i == 0) {
                isPrime = 0;
                break;
            }
        }

        if (isPrime) {
            printf("%d ", num);
        }
    }
    printf("\n");
}
int main() {
    int limit;
    printf("Enter the limit: ");
    scanf("%d", &limit);
```

```
    printf("Prime numbers up to %d are: ", limit);
    findPrimeNumbers(limit);

    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/ou
./"q18"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q18"
Enter the limit: 5
Prime numbers up to 5 are: 2 3 5
tenzinlhamo@TENTENs-MacBook-Pro output %
```

19. Write a function named calculateFactorialSeries that takes an integer n as input and prints the factorial series up to n.

Code

```
#include <stdio.h>
void calculateFactorialSeries(int n) {
    if (n < 1) {
        printf("Please enter a positive integer.\n");
        return;
    }

    long long fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
        printf("%d! = %lld\n", i, fact);
    }
}

int main() {
    int n;
    printf("Enter the value of n: ");
    scanf("%d", &n);

    printf("Factorial series up to %d:\n", n);
    calculateFactorialSeries(n);
    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q19"
Enter the value of n: 6
Factorial series up to 6:
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
6! = 720
tenzinlhamo@TENTENs-MacBook-Pro output %
```

20.Write a function named calculateGCD that takes two integers as input and returns their greatest common divisor (GCD).

Code

```c
#include <stdio.h>
int calculateGCD(int a, int b) {
   int temp;
   while (b != 0) {
       temp = b;
       b = a % b;
       a = temp;
   }
   return a;
}

int main() {
   int num1 = 48, num2 = 18;
   printf("The GCD of %d and %d is: %d\n", num1, num2, calculateGCD(num1, num2));
   return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q20"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q20"
The GCD of 48 and 18 is: 6
tenzinlhamo@TENTENs-MacBook-Pro output %
```

21.Write a recursive function named calculateFactorial that takes an integer n as input and returns its factorial.

Code

```c
#include <stdio.h>
long long calculateFactorial(int n) {
   if (n <= 1) {
       return 1;
   }
   return n * calculateFactorial(n - 1);
}

int main() {
   int number = 5;
   printf("Factorial of %d is %lld\n", number, calculateFactorial(number));
   return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q21"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q21"
Factorial of 5 is 120
tenzinlhamo@TENTENs-MacBook-Pro output % 5
```

22. Write a recursive function named calculateFibonacci that takes an integer n as input and returns the nth Fibonacci number. The Fibonacci sequence starts with 0 and 1, and each subsequent number is the sum of the two preceding numbers.

Code

```c
#include <stdio.h>
long long memo[100];

long long calculateFibonacci(int n) {
   if (n <= 0) return 0;
   if (n == 1) return 1;
   if (memo[n] != 0) {
       return memo[n];
   }
   memo[n] = calculateFibonacci(n - 1) + calculateFibonacci(n - 2);
   return memo[n];
}
int main() {
```

```
    int n = 50;
    printf("The %dth Fibonacci number is: %lld\n", n, calculateFibonacci(n));
    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q22"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q22"
The 50th Fibonacci number is: 12586269025
tenzinlhamo@TENTENs-MacBook-Pro output % 5
```

23.Write a recursive function named calculateGCD that takes two integers a and b as input and returns their greatest common divisor (GCD).

Code

```
#include <stdio.h>
int calculateGCD(int a, int b) {

    if (b == 0) {
        return a;
    }
    return calculateGCD(b, a % b);
}

int main() {
    int num1 = 48, num2 = 18;
    printf("The GCD of %d and %d is: %d\n", num1, num2, calculateGCD(num1, num2));
    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q23"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q23"
The GCD of 48 and 18 is: 6
tenzinlhamo@TENTENs-MacBook-Pro output %
```

24.Write a recursive function named calculatePower that takes two integers base and exponent as input and returns the result of raising the base to the exponent.

Code

```
#include <stdio.h>
long long calculatePower(int base, int exponent) {
   if (exponent == 0) {
       return 1;
   }
   return base * calculatePower(base, exponent - 1);
}
int main() {
   int b = 2, e = 10;
   printf("%d raised to the power of %d is %lld\n", b, e, calculatePower(b, e));
   return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q24"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q24"
2 raised to the power of 10 is 1024
tenzinlhamo@TENTENs-MacBook-Pro output %
```

25.Write a recursive function named calculateSumOfDigits that takes an integer n as input and returns the sum of its digits.

Code

```
#include <stdio.h>
int calculateSumOfDigits(int n) {
   if (n == 0) {
       return 0;
   }

   return (n % 10) + calculateSumOfDigits(n / 10);
}

int main() {
   int num = 1234;
   printf("Sum of digits of %d is %d\n", num, calculateSumOfDigits(num));
   return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q25"
Sum of digits of 1234 is 10
tenzinlhamo@TENTENs-MacBook-Pro output %
```

26. Write a recursive function named reverseString that takes a string as input and returns the reversed string.

Code

```c
#include <stdio.h>
#include <string.h>
void reverseString(char *str) {

   if (*str == '\0') {
       return;
   }
   reverseString(str + 1);
   printf("%c", *str);
}

int main() {
   char str[] = "Hello World";
   printf("Original: %s\n", str);
   printf("Reversed: ");
   reverseString(str);
   printf("\n");
   return 0;
}
```

Output

```
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
./"q26"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q26"
Original: Hello World
Reversed: dlroW olleH
tenzinlhamo@TENTENs-MacBook-Pro output %
```

27.Write a recursive function named printTriangle that takes an integer n as input and prints a triangle of asterisks (*) with n rows.

Code

```c
#include <stdio.h>
void printRow(int m) {
   if (m == 0) {
       return;
   }
   printf("* ");
   printRow(m - 1);
}
void printTriangle(int n) {
    if (n <= 0) {
       return;
   }
   printTriangle(n - 1);
   printRow(n);
   printf("\n");
}

int main() {
   int rows = 5;
   printTriangle(rows);
   return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q27"
*
* *
* * *
* * * *
* * * * *
tenzinlhamo@TENTENs-MacBook-Pro output %
```

28.Write a recursive function named calculateBinary that takes an integer n as input and returns its binary representation as a string.

Code

```c
#include <stdio.h>
#include <string.h>
void convertToBinary(int n, char *buffer, int *index) {
```

```
    if (n > 1) {
        convertToBinary(n / 2, buffer, index);
    }

    buffer[(*index)++] = (n % 2) + '0';
    buffer[*index] = '\0';
}

int main() {
    int n = 13;
    char binaryString[33];
    int index = 0;

    if (n == 0) {
        strcpy(binaryString, "0");
    } else {
        convertToBinary(n, binaryString, &index);
    }

    printf("Decimal: %d\nBinary: %s\n", n, binaryString);
    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL
tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q28"
Decimal: 13
Binary: 1101
tenzinlhamo@TENTENs-MacBook-Pro output %
```

29.Write a recursive function named isPalindrome that takes a string as input and returns 1 if it is a palindrome (reads the same forwards and backwards), and 0 otherwise.

Code

```
#include <stdio.h>
#include <string.h>
int checkPalindrome(char *str, int left, int right) {
    if (left >= right) {
        return 1;
    }
```

```c
    if (str[left] != str[right]) {
        return 0;
    }
    return checkPalindrome(str, left + 1, right - 1);
}
int isPalindrome(char *str) {
    int len = strlen(str);
    if (len == 0) return 1;
    return checkPalindrome(str, 0, len - 1);
}
int main() {
    char str[] = "radar";
    if (isPalindrome(str)) {
        printf("'%s' is a palindrome.\n", str);
    } else {
        printf("'%s' is not a palindrome.\n", str);
    }
    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q29"
'radar' is a palindrome.
tenzinlhamo@TENTENs-MacBook-Pro output % %
```

30. Write a recursive function named countOccurrences that takes a string and a character as input and returns the number of times the character appears in the string.

Code

```c
#include <stdio.h>
int countOccurrences(char *str, char target) {

    if (*str == '\0') {
        return 0;
    }
    if (*str == target) {

        return 1 + countOccurrences(str + 1, target);
    } else {
        return countOccurrences(str + 1, target);
    }
}
int main() {
```

```c
    char myString[] = "banana";
    char searchChar = 'a';

    int result = countOccurrences(myString, searchChar);

    printf("The character '%c' appears %d times in \"%s\".\n",
            searchChar, result, myString);

    return 0;
}
```

Output

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

TERMINAL

tenzinlhamo@TENTENs-MacBook-Pro output % cd "/Users/Dolkar/Documents/CPrograming /CProgramming/assignment-3-tdolkar13/src/output"
tenzinlhamo@TENTENs-MacBook-Pro output % ./"q30"
The character 'a' appears 3 times in "banana".
tenzinlhamo@TENTENs-MacBook-Pro output %
```